

Important aspects of implementing a concurrent design and test approach

Appropriate tools are required during the design process in order to facilitate the concurrent design and test approach. By the expression 'concurrent design and test,' what is being referred to is a design methodology where simulation, verification, hardware implementation and software development need to happen with close feedback from each other. In such a scenario, the tools used for each of the above tasks should gel seamlessly with each other. Otherwise, there is a risk of engineers spending more time in getting the tools to work rather than getting the job done. Let us look further into what aspects of an ecosystem are important for implementing the concurrent design and test approach.

1. Co-simulation between multiple simulation environments. Engineers should have access to simulation tools that facilitate system/subsystem simulation. In many cases, a single simulation tool may not be capable of simulating the entire system. Hence, it is very important to have the appropriate simulation environments as well as seamless integration between these simulation tools.

2. Powerful hardware targets and quick algorithm deployment. Once simulated, the ecosystem needs to provide powerful and versatile hardware targets to deploy the code on. The deployment of the algorithm developed should not be a tedious task in itself. This requires that the software used for algorithm development should easily integrate with the hardware platform.

3. 'Hardware in the Loop' testing of design. As the subsystems are built, they need to be tested assuming that the other subsystems are already available. This calls for 'hardware in the loop' testing (HIL) of the subsystem. To implement such tests, the hardware should be capable of real-time simulation and deterministic I/O.

4. Simulation and test software connectivity. Simulation software can be thought of as a utility that uses mathematical models to represent the output of a device under test (DUT) based on a series of inputs, while the actual verification and validation (V&V) involves a prototype DUT. Connectivity between design and test software environments allows design engineers to utilise significantly richer measurement algorithms earlier in the design process. It also allows easy reuse of test procedures between simulation and V&V stages.

—Contributed by: **Raviteja Chivukula, technical marketing engineer, National Instruments**

gregation," explains Stephen Hire. As the design progresses, they will then use the test mobile to help them debug any issues and to help optimise critical performance elements such as scheduling algorithms and data throughput.

Once the design phase is completed and the prototype base station is ready, it will usually be handed over to a different group for conformance testing to ensure that the base station meets all of the relevant 3GPP standards — this latter stage can be considered as 'design after testing.' Another aspect of testing after design comes in whenever a new version of software is released. For most commercial products, there is usually a lot of regression testing to ensure that all the features from earlier versions of software still work correctly with the latest version.

This design methodology involves product design happening hand-in-hand with test benches for the product.

"In the automotive and aerospace industries, it has been practised with varying degrees of success in the form of the V-diagram model. While the use of concurrent design and test processes such as the V-diagram is common and obvious in these industries, adoption of these practices is growing in other industries and for other types of devices as well. For example, in the semiconductor and consumer electronics industries, shorter product lifespans and increasing product complexity continually drive pressure to reduce product development time by adopting the 'test while design' strategy," explains Chivukula.

Traditionally, the left side of the V-diagram has been viewed as an R&D or design task and the right as test and validation tasks, and these two phases of product development have been separated. Once the production control unit is manufactured, it is passed to the

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